Delft University of Technology
Department of Control & Simulation

Progress Report A. de Windt

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Date:	07-02-2019
Subject:	Policy Gradient Reinforcement Learning for quadrotor control
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Project term:	09/2018 - 07/2019
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Outcome of the previous meeting

- Didn't do much progress.

Description of the results since last meeting

Short version

- Pole example with tabular Q-learning working.
- n-step bootstrapping implemented, but it's seems to make things worse.
- Very rough outline of literature study report.

Plans for next 2 weeks

1. Implement RL agent with value function approximation (eg. tile-coding or fourier)
2. Focus more on policy gradient methods. I didn't understand these initially, so I have not looked much into this.
3. Write the literature study report.

A Simulation results

I managed to get a tabular Q-learning RL agent to control the pole. The results shown below are for an agent with 1681 different states (41 θ and 41 $\dot{\theta}$) and 41 different actions. So 68921 different state-action combinations. It's possible to get the agent working just fine with considerably less state-value combinations and simulations. These plots just look nicer.

To train the agent I first ran 40000, 1 second simulations with $\epsilon = 0.7$, $\gamma = 0.7$ and constant $\alpha = 0.9$. On the second batch of 5000 cases I increased ϵ to 0.9. On the last batch I increased ϵ to 1.0 and decreased α to 0.0 in order to see the ideal policy of the agent without any learning.

Figure 1 shows the ideal action for each state. As it can be seen the agent is showing the expected behaviour. However it doesn't seem like it's learning properly when $\dot{\theta} \approx 0$.

- It applies a positive torque when θ is negative and vice-versa.
- It chooses a lower torque if the pole is already moving towards the center and increases the torque if it's moving away.

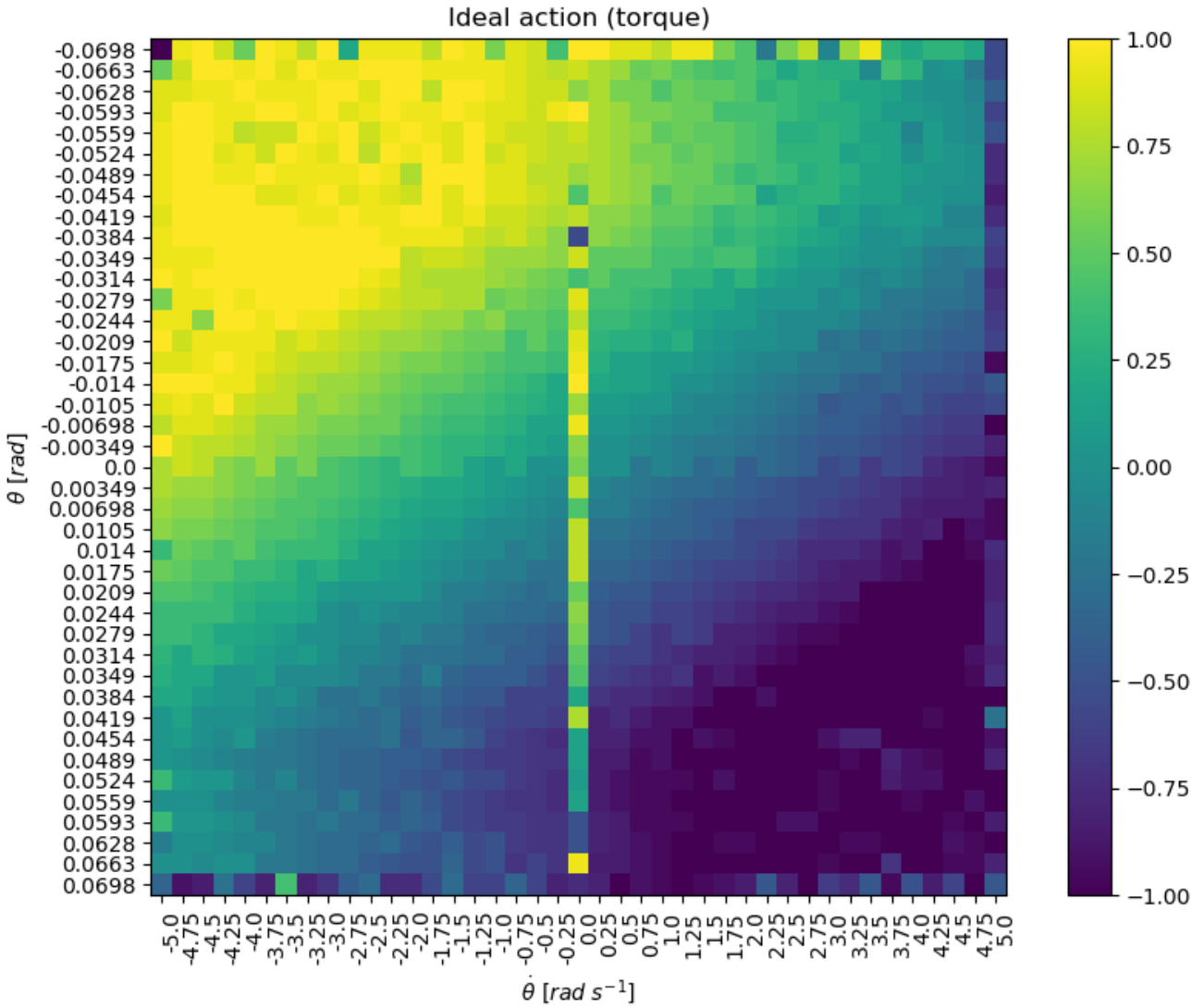

Figure 1: Ideal action at each state.

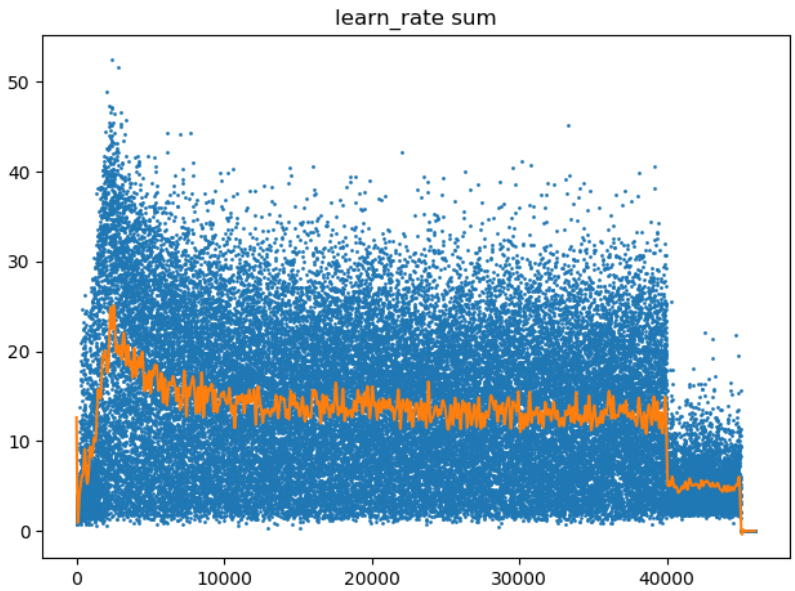

Figure 2: The sum of the changes made to the value table on each episode.

Figure 3: The total reward sum for each episode.

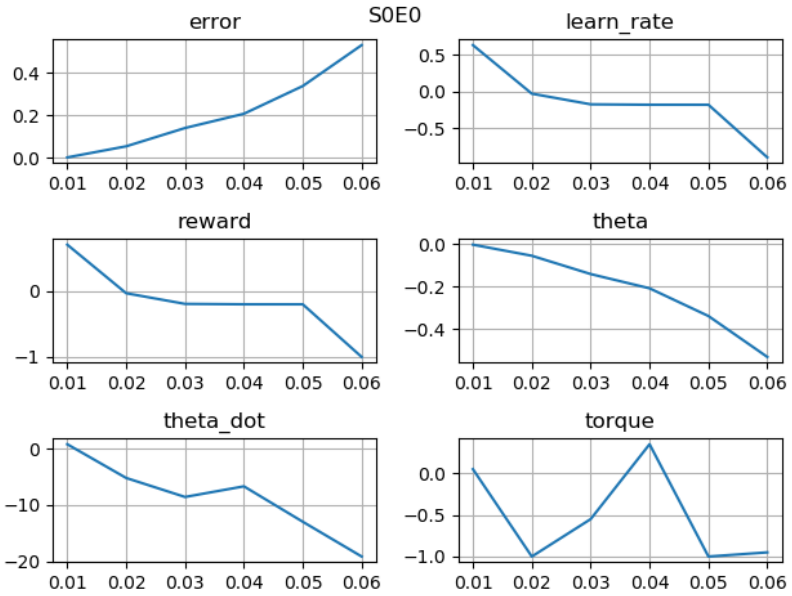

Figure 4: Simulation result of the first case ($\epsilon = 0.7$, $\gamma = 0.7$ and $\alpha = 0.9$).

Figure 5: Simulation result of the last case of the first batch ($\epsilon = 0.7$, $\gamma = 0.7$ and $\alpha = 0.9$).

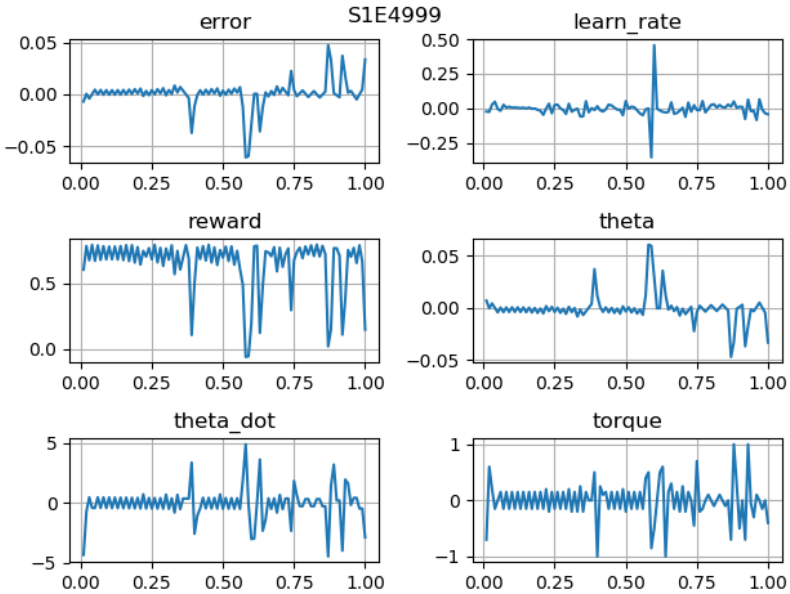

Figure 6: Simulation result of the last case of the second batch ($\epsilon = 0.9$, $\gamma = 0.7$ and $\alpha = 0.9$).

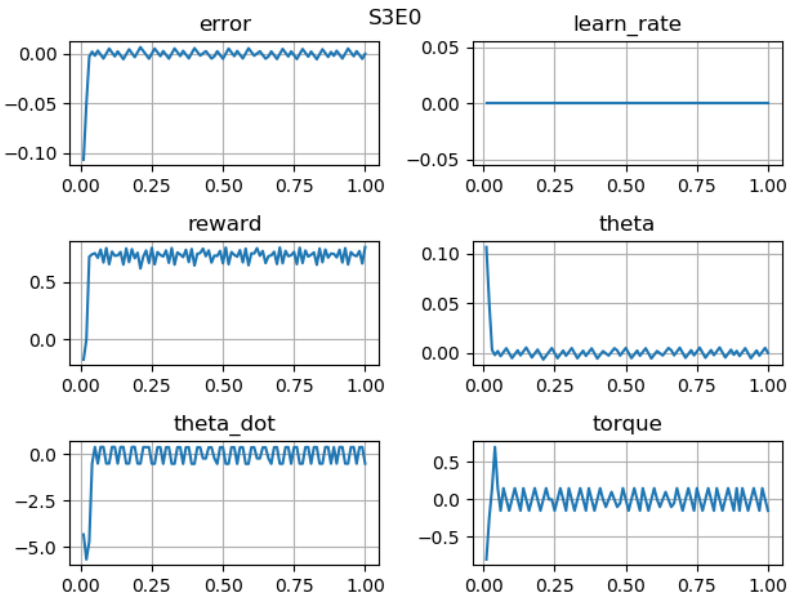

Figure 7: Simulation result of the ideal policy without any learning ($\epsilon = 1.0$, $\gamma = 0.7$ and $\alpha = 0$).